Stage 1: Build Sparse 4D Object Tracks from Stereoscopic Videos (Data Pipeline)

1. Stereo video clips (Stereo4D)



monocular / stereo video (e.g., Stereo4D)

2. Object inventory (Qwen3-VL-Plus)

- person ×12
- bicycle
- car
- storefront
- …

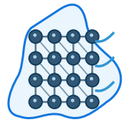
3. Instance masks (SAM 3)



…

4. 3D tracking (Track4World or GT)

Track4World (monocular video)



dense tracking
+ world frame
+ camera poses

or

GT 3D tracks (e.g., Stereo4D)



assign tracks to masks
visibility check
moving ▶ static

5. Object geometry (mesh ▶ parts ▶ keypoints)

non-rigid



rigid — 4 points + SE(3)

static — ≤ 4 points

6. Sparse 4D object tracks



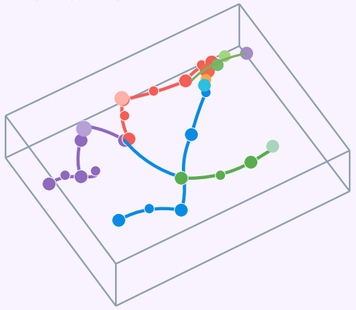
~45 keypoints / clip (world frame)

Stereo4D GT-track branch: 108k clips ▶ 42k kept

filters: moving object · valid mask · ≥ 8 tracks · motion check

Stage 2: Trajectory Tokenization and VLM Training (Model Pipeline)

1. Trajectory tokenization (Fig. 3)



sparse 4D tracks ▶ trajectory tokens

2. Model inputs



8 video frames

trajectory tokens (+ object / part text)

question / instruction (motion captions, trajectory QA)

3. VLM with trajectory tokens

Vision encoder (frozen)

Trajectory Token Projector (trainable)

LoRA

Qwen3-VL-8B

Text tokens (question, captions)

4. Training strategy

Projector warm-up ▶ **Token-only** fine-tuning ▶ **Joint** fine-tuning

Supervision from 4D ground truth

trajectory QA
- object displacement
- average speed
- camera displacement
- motion duration
- …

motion captions
"The bicycle moves about 13.5 m in the world frame at an average speed of 4.58 m/s …"

5. Qualitative example

Q: In the world frame, how far did the backpack move between frames 4 and 14?

GT: 2.09 m

Frames only — 1.1 m ✗

Frames + trajectory tokens — 2.2 m ✓